

To: Divyanshu

From: Rohit

Re: Business type expansion (banks + high margin hardware) — answers to your 11 questions

Hey — went through all 11 of your questions below. We already have 4 business types built and specced — saas, compounder, income, cyclical — nothing needed there, all parameters are clear in current code.

We're adding 2 more, not the 5 you listed: banks and high margin hardware. Not insurers, not REITs, not biotech, not deep value.

- REITs — ditch them, don't care about them.
- Biotech — don't care, apart from IBB, and IBB is an ETF, not a single-stock name, so it never reaches business-type logic anyway — it gets caught by the Step 0 asset-type gate and returns NOT_APPLICABLE before BQ scoring runs. Nothing to build there.
- Insurers and deep value — default to the lighter option: coverage_incomplete flag only (see Q4), no full valuation logic yet.

One thing to clear up first: "high margin software" isn't a missing category — that's already the existing saas type via the Rule of 40 test. The new hardware bucket is purely for semiconductor/hardware names that don't behave like a normal cyclical business but also aren't SaaS, income, or compounder — NVDA being the trigger case.

Q1 — How much of the business type work do we build

Decision: build banks fully. Build high_margin_hardware fully. Insurers and deep value get the coverage_incomplete flag only and post this project, we can look at that if/when we work together. REITs and biotech are dropped from this round entirely.

Banks doesn't map onto any of the 4 existing type substitutions — no FCF statement the same way, no EV/Revenue multiple anyone actually uses, and "debt" is deposits so net debt/EBITDA is meaningless. Here's the parallel calibration, sourced entirely from data we're already pulling via yfinance — no new vendor:

Parameter	Banks (new)	Source
Detection	sector = Financial Services AND industry contains Bank (Commercial/Regional/Diversified Banks)	yfinance info sector/industry — same fields already used for the other 4 types
margin_quality substitution	Efficiency ratio = noninterest expense / (net interest income + noninterest income). <55% = +2, 55-65% = 0, >65% = -2	income_stmt — already-pulled dataset
balance_sheet substitution	Equity/assets ratio as leverage proxy. >10% well-capitalized, 6-10% adequate, <6% flag	balance_sheet — already-pulled dataset
roic_wacc_spread substitution	ROE vs ~9-10% cost of equity	info returnOnEquity — already-pulled
OEY substitution	Earnings yield = Net Income / Market Cap (1/PE)	income_stmt net income + market cap — already-pulled
valuation_tax substitution	Price/Book tiers: Tier1 >=1.5x, Tier2 >=2.0x, Tier3 >=2.5x, floor >=3.0x	info priceToBook — already-pulled
WACC	9%	New constant, same pattern as existing 4 types

yield_trap market-absolute threshold	+2pp on top of existing threshold for this type (e.g. US/India 6% -> 8%)	Same z-score + threshold logic, adjusted constant
PE 20Y percentile	No change — already self-calibrating per ticker	n/a

Q2 — Margin cutoff for high margin hardware

40% trailing net margin, computed off TTM (most recent 4 quarters), refreshed every quarterly recalculation — not a one-time static classification. Cleanly separates genuinely exceptional-margin hardware/semi names from the broader group without only catching one ticker.

We stop using G2 as evidence for hardware/semi names' competitive_moat score, since it measures software UX satisfaction, not chip competitiveness.

Q3 — Should G2 exclusion follow the new bucket or the raw sector

Raw sector, not the new bucket. G2 is equally irrelevant to a 25%-margin chip company as a 55%-margin one. Key the exclusion off sector/industry directly, at the same point business_type detection reads those fields.

For hardware names below the 40% threshold (still classified cyclical by default): competitive_moat sourcing swaps to Gartner + patent databases — same non-G2 stack, without the rest of the high_margin_hardware calibration (WACC, valuation tiers, margin_quality stay on cyclical defaults for those names).

Q4 — What happens to the verdict when coverage is incomplete

Hard gate, and its own verdict. Build COVERAGE INCOMPLETE as a third hard gate alongside yield_trap and FS=weak, checked at the same point in the Layer D signal modifier. Sizing forced to 0% either way, but the label matters: CANCEL BUY means "we evaluated this and the answer is no." COVERAGE INCOMPLETE means "we haven't built a reliable way to score this business type yet." Those are different messages should look different on the dashboard.

This fires whenever business_type detection can't map a ticker to one of the calibrated types (saas/compounder/income/cyclical/bank/high_margin_hardware) — instead of letting it silently fall through to the cyclical default the way it does today.

Yes, this needs a new verdict case on Parth's side (color/label logic currently only covers Green=BUY / Amber=REDUCED-CANCEL / Red=YIELD TRAP) — go ahead and loop him in for it, it's a small addition.

Q5 — What counts as a buyback suspension

Scaled, not binary. Build as a standalone penalty flag, structurally separate from mgmt_capital_allocation — same pattern as how Altman Z is handled as a penalty flag rather than folded into balance_sheet, to avoid double-counting.

Trigger condition: prior-period buyback spend must exceed \$100M.

Buyback spend decline (period-over-period)	Recalc trigger?	Penalty (standalone flag)
0-25%	No	0
25-50%	Yes	-1

50-75%	Yes	-2
75-100% (near/full suspension)	Yes	-3

This is a spend-based trigger, distinct from the existing share-count-based buyback vote inside `fd_direction` — that vote should keep running independently.

Q6 — What size dividend cut should trigger a recalc

Same logic as buyback — same tier boundaries, same standalone-flag structure, applied to declared annual DPS:

Dividend per share decline (vs. prior declared annual rate)	Recalc trigger?	Penalty (standalone flag)
0-25%	No	0
25-50%	Yes	-1
50-75%	Yes	-2
75-100% (elimination)	Yes	-3

If both the buyback and dividend flags fire in the same quarter, cap the combined penalty from these two flags at -4 total rather than letting them stack to -6.

Q7 — Where does the revenue miss trigger get its expected number from

Use yfinance's analyst revenue estimate field — same integration pattern as the EPS estimate already wired for the `fd_direction` EPS-revisions vote, so no new data source or key needed.

Trigger `full_recalculation()` when actual revenue **misses the stored consensus estimate** by >10%. No new scoring dimension needed for this — `growth_trajectory` and the Revenue vote inside `fd_direction` already pick up the consequence automatically once recalculation runs, so don't build a separate penalty on top of it.

Q8 — Does the supply constraint flag move the score at all

Information only, no automatic score change. Store it in the rationale string the same way TAM sourcing is already cited today. No build needed beyond making sure the flag text lands in the stored rationale.

Q9 — What tax rate do we use for the adjusted EPS work

Use the company's own trailing effective tax rate — not a flat 21% or 25%. Trailing 4Q tax provision ÷ trailing 4Q pretax income, both already available in the `income_stmt` pull we're already making for every other BQ dimension. Zero new data source.

Q10 — Should adjusted PE always replace raw PE or only sometimes

Materiality gate: substitute adjusted PE for raw PE only when one-off items exceed 5% (as you suggested Divyanshu) of trailing net income. Otherwise leave raw PE in the percentile calculation untouched.

Q11 — How much of the universe gets rerun once this is built

Run a cheap classification-only pass first — just the business_type auto-detect logic (sector/industry read, no full financials pull) — across all approx. 193 tickers. Same cost profile as the daily lightweight runner, not a full Layer C pull.

Diff against each ticker's currently stored business_type. Only the ones that actually flip (into bank, high_margin_hardware, or coverage_incomplete) get queued for a full full_recalculation(). Everything that classifies the same as before stays on its normal schedule. Same migration discipline as the PE history rollout, without paying for a blanket rerun.

Let me know once you've got a rough sense of how many tickers in the 193 flip into banks/hardware, so we know the real size of the recalculation queue in Q11.

Live dashboard CLAUDE Sonnet 5 audit — CRM conviction mismatch + other findings 🔗

Pulled up the full Signal status & depth page (not just the CRM drawer). One real bug worth chasing, plus three smaller UI items.

CRM conviction number doesn't match its own breakdown, on this exact screen.

Top bar and the signal chip both show CRM +1 · CANCEL. But the drawer right below shows BQ +8, Tax -5 → that's +3, not +1. +3 should land as REDUCED BUY ($\geq +2$), not CANCEL.

Specific hypothesis worth testing: per the FS cap table, the weak-tier cap on a long signal is $\min(\text{conviction}, +1)$ — and +1 is exactly what's displayed. Either fs_class is actually reading as weak somewhere in the cap-time pipeline while the drawer's "Strong" label is pulling a different (possibly stale) fs_class value, or the cap logic is hard-applying the weak-tier formula regardless of the real fs_class. Please print both the fs_class used at cap-time and the one rendered in the UI for CRM specifically — if they don't match, that's the bug. Same class of gap-testing that caught the PYPL issue.

- Sidebar copy is stale: "BQ Score / 15 dims / 7 auto + 8 manual" in the left nav contradicts every dimension card on the page, which all show AUTO. Leftover from the pre-"all 15 auto" era — quick fix for Parth, no scoring impact.
- Yield Traps panel reads as contradictory: big number says 0, text underneath says "T.TO, WIPRO.NS — threshold breached." If those two cleared the 2-condition hard gate, the count should say 2. If it's a soft/near-threshold warning, the label needs to say that explicitly (e.g. "0 confirmed / 2 watching") instead of looking like a bug.
- Business Types panel shows "Unknown · 1" — live example of exactly what the coverage_incomplete gate (Q4) is meant to catch. Once that gate's built, this ticker is the built-in test case.

Open items before this is fully defensible

Went back through the v5 spec, the Fundamental Agent Update, and the live signal universe on the dashboard specifically to check whether you have everything you need for a defensible FS score built from BQ raw and valuation tax? Short answer: not quite — one real formula gap, plus three sourcing gaps that show up concretely in tickers already on the board.

1. The fs_score "daily valuation slice" formula is never fully specified.

fs_quality_base = $50 + (\text{BQ_RAW} \times 2.5)$ is fully defined. But fs_score = fs_quality_base + daily valuation slice (OEY, PE percentile, EV/rev) has no general point-value table anywhere — the only numeric example is the TELUS case ("OEY strong $\geq 3\%$ → +5"), and even that doesn't show the PE-percentile or EV/rev contribution explicitly. Since

fs_score directly caps conviction_score, this is the single biggest gap standing between what's spec'd and a formula we can actually defend if asked how a number was produced. Needs the same kind of explicit tier table already built for valuation_tax (Section 5.4), just for the fs_score slice.

2. CEO tenure sourcing is US-only, but the live universe isn't.

ceo_start_date is spec'd as sourced from SEC DEF14A — a US filing type. WIPRO.NS, BPCL.NS, PERSISTENT.NS, 005930.KS, and 000660.KS are all live on the dashboard right now, and none of them file a DEF14A. No equivalent source named for Canadian management circulars, Indian exchange disclosures, or Korean filings. Use Tavela?

3. Insider ownership sourcing has the same gap.

Spec'd as SEC EDGAR (US) / SEDI (Canada) only. The Indian and Korean names above have no named source either. Use Tavela?

4. PE 20Y history for non-US tickers — automated three-tier fallback, no manual step.

Fix: Tier 1 — try `fetch_pe_history_macrotrends()` for every ticker, not just US ones (drop the US-only gate, let it fail naturally if the ticker isn't covered).

Tier 2 — if that fails, build the PE array in code: pull price history via `stock.history()` and trailing-4Q EPS at each historical point from `quarterly_income_stmt` (same net-income/shares fallback already built for the PYPL bug, just run across the full available window instead of just TTM), divide to construct a synthetic PE series.

Tier 3 — if both come up empty (young listing, thin financials), flag `pe_history_thin` with a stored `pe_history_years` count rather than blocking the ticker or falling back to a manual edit.

Realistic caveat: Tier 2 will typically only reach 5–8 years of depth for most Indian/Korean names, since yfinance's quarterly financials don't go back 20 years even where price history does — shorter than the US path, but automated and honestly labeled via `pe_history_years`, instead of a manual step nobody will actually sit down and do. True 20-year depth to match the US path would need a paid Gurufocus API subscription.

5. yield_trap market-absolute thresholds don't cover Korea, Japan, or China — deprioritized, not currently load-bearing.

005930.KS, 000660.KS, SFTBY, and BABA are the only KR/JP/CN names on the current signal list, and none of them are dividend-yield plays — they're growth or growth-stalled names, not the profile yield_trap exists to catch. So the missing thresholds aren't blocking anything in the live portfolio today. Still worth you having a placeholder number ready for whenever a genuine dividend payer from those markets enters the universe, but unlike the other four items, this one isn't a pre-production-ready blocker.

For Parth — Engine Layers click-through views

The three static Engine Layers entries in the left sidebar (Valuation Tax · sector-calibrated, FS Cap · graduated sizing, Yield Trap · 2-condition hard gate) don't do anything when clicked today. Proposed content for each, scoped to whichever ticker is currently selected (falls back to a general explainer if none is selected). All three need a `coverage_incomplete` empty state — for a ticker gated there, show "not yet calibrated for this business type" instead of blank or zeroed fields.

Valuation Tax view

Component-by-component ledger for the selected ticker, branched by `business_type`: for `saas/compounder/income/cyclical/high_margin/hardware` — `entry_multiple` tier hit, `pe_hist_percentile` penalty, `growth_multiple_fragility`, `business_type` relief, floor trigger status. For banks — Price/Book tier hit in place of `entry_multiple`, same PE-percentile and floor-trigger rows. Each row shows the raw input, the tier boundary it's compared against, and the resulting point deduction, summing to the

displayed TAX value. Same data already spec'd for the drawer's "valuation tax breakdown text" — just promoted to its own page.

FS Cap view

fs_quality_base, each daily valuation slice input branched by type (OEY vs. threshold, or for banks earnings yield vs. threshold; PE percentile vs. bucket, shown with pe_history_years alongside it so a 6-year read isn't mistaken for a 20-year one; EV/fwd rev vs. tier, or for banks P/B vs. tier), the resulting fs_class, and which cap — if any — was applied to conviction_score, shown side by side with the long-signal and short-signal cap from the Section 4.3 table. This view is what would have caught the CRM +1-vs-+3 mismatch immediately.

Yield Trap view

Both conditions shown explicitly with pass/fail state: dividend z-score vs. 5Y own history (actual number vs. the >1.5 threshold), and yield vs. the market-absolute threshold — shown as base country threshold + any business-type adjustment (e.g. bank +2pp) = the actual number applied. For tickers in a market with no threshold defined yet (currently Korea/Japan/China), show that explicitly rather than comparing against nothing. Both conditions must be true to fire. This directly resolves the Yield Traps panel ambiguity — it would show T.TO and WIPRO.NS's exact condition-by-condition status. Forwarding this section to you so you can route it to Parth directly, or I can send it separately — let me know which you'd rather do.